Minimax Algo.

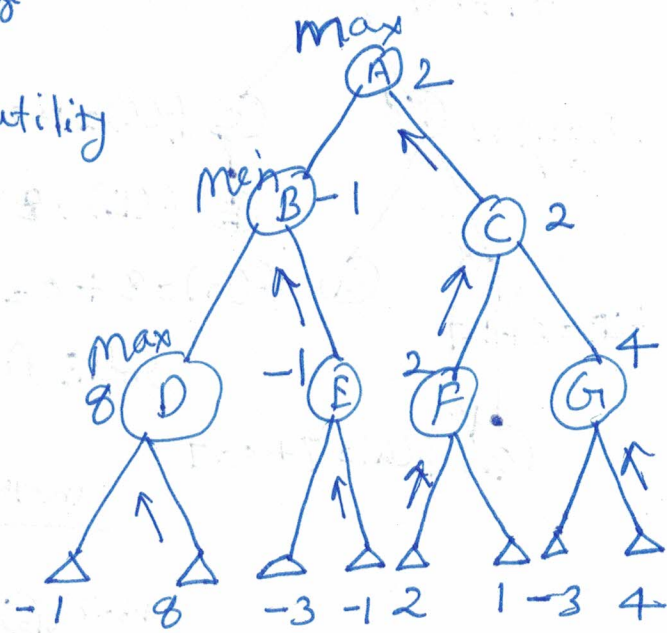
Goal : find optimal path that can maximize the utility.

- Back tracking algorithm
- Best move strategy used
- Max will trying to maximize its own utility (Best move)
- Min will try to minimize utility (worst move) of opponent

$O(b^d)$

b = branching factor

d = depth

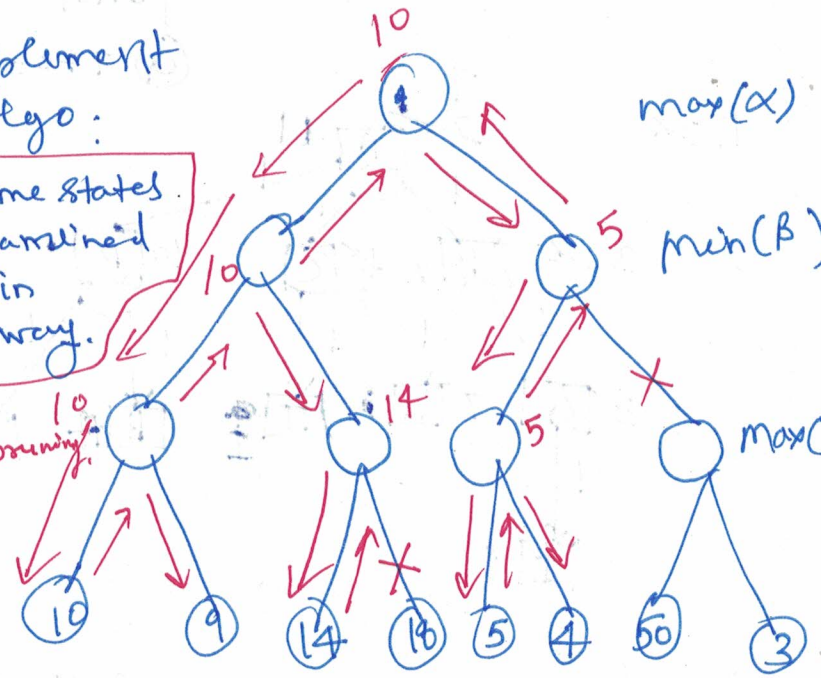


Game tree

α-β pruning Alg : Implement (Alpha-Beta) DFS Algo :

Why α-β pruning : The no. of game states has to be examined which may go in exponential way.

Compare : $\alpha \geq \beta$



- min is not having any value so visit (9). (β)
- there is the single value at α i.e. 10, so propogate that value to node B.

* pruned edge

Constraint Satisfaction Problem (CSP)

* Many AI problems can be viewed as part of constraint satisfaction, in which the goal is to discover some problem state that satisfies a given set of constraints.

  - Graph colouring problems → Given a graph of vertices and edges, you want to colour the nodes in such a way that no two adjacent vertices share a colour using the fewest no. of colours
  - Cryptarithmetic puzzles ↳ same digit cannot be assigned to two letters.
  - Sudoku game - each digit (1-9) must appear exactly once in every row, column, and (3x3) subgrid

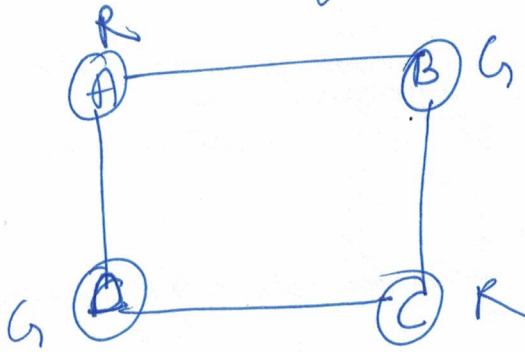
CSP has three components:

V is the set of variables $\{V_1, V_2, V_3, \ldots V_n\}$

D is the set of domains $\{D_1, D_2, D_3\}$ one for each variable

C is the set of constraints that specify allowable combination of values.

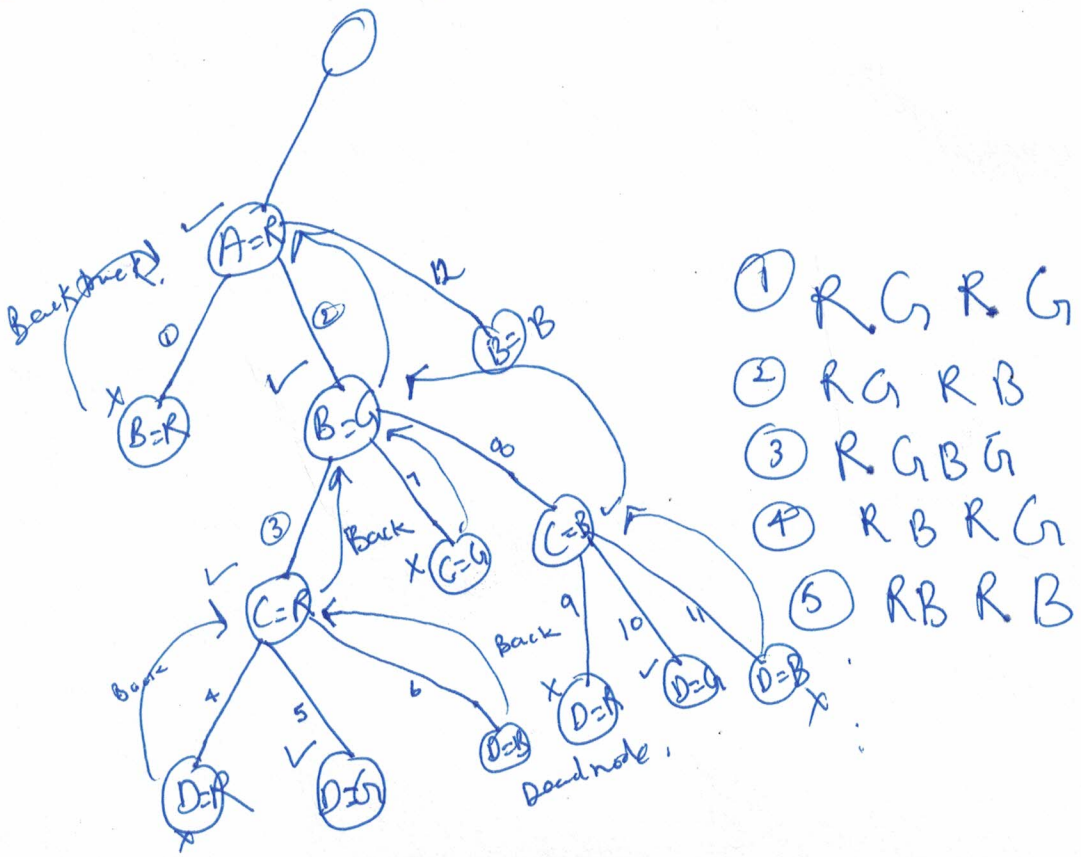
Backtracking - It gives multiple solutions..



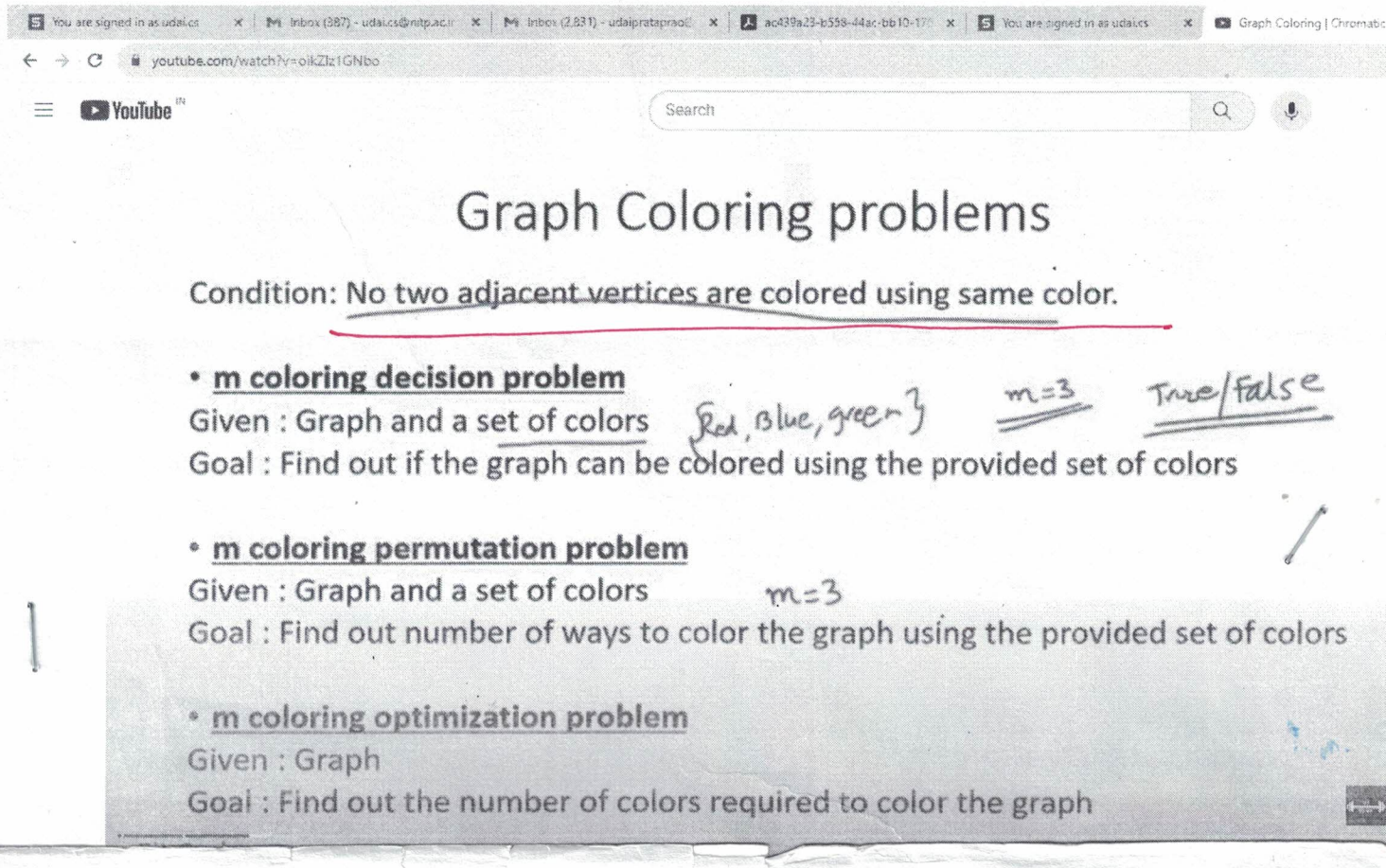
{R, G, B}

→ Graph Colouring Problem

Generate State Space tree



① R G R G
② R G R B
③ R G B G
④ R B R G
⑤ R B R B


youtube.com/watch?v=oikZIz1GNbo
YouTube
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Graph Coloring problems
Condition: No two adjacent vertices are colored using same color.
• m coloring decision problem
Given : Graph and a set of colors
Goal : Find out if the graph can be colored using the provided set of colors
• m coloring permutation problem
Given : Graph and a set of colors
Goal : Find out number of ways to color the graph using the provided set of colors
• m coloring optimization problem
Given : Graph
Goal : Find out the number of colors required to color the graph

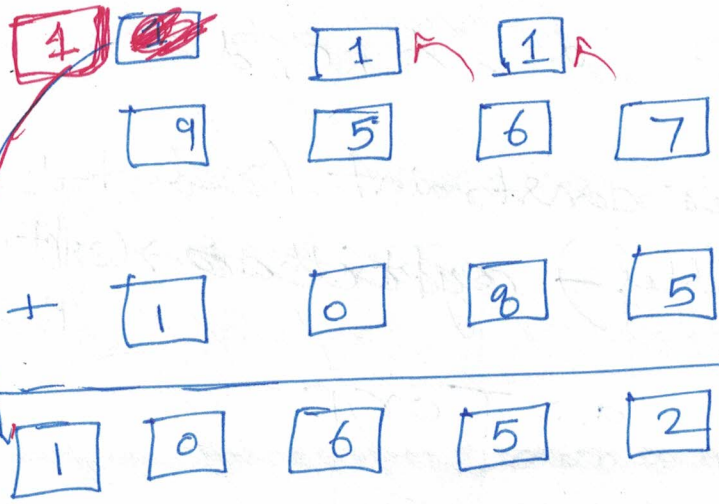
Cryptarithmetic Problem → Whenever we are trying to transmit any particular message, we need to encrypted in such a manner that it is not readable to any person to whom we do not intent to read

```
  S E N D
+ M O R E
---------
M O N E Y
```

{ Domains: {0, 1, 2, 3, 4, 5, 6, 7, 8, 9}
Same domain for all variables }

Constraint: We cannot assign the same digit to two letters.

$M \neq 0$



S — 9
E — 5
N — 6
D — 7
M — 1
O — 0
R — 8
Y — 2

sequence: M, S, E, N

this may not be unique solution, there might be possibility, you can use diff. possible combinations.

Solve the following expression

```
  B A S E
+ B A L L
---------
G A M E S
```

Add. Constraints.

$G \neq 0$


```
  TWO
+ TWO
-----
 FOUR
```

Variables: F, T, U, W, R, O
Domains: {0, 1, 2, 3, 4, 5, 6, 7, 8, 9}
Constraints:
$F \neq T$, $F \neq U$, $F \neq 0$
multivariable constraints: $O + O = R + 10$